

ZEERBOX · REQUIREMENTS

ZeerBox requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Status from ZBX-CAL-001 v0.3 after the design for construction (ZBX-DDR-003); R12 reported against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R3 relaxed to 80 %; status from ZBX-CAL-001 v0.2 (150 mm pad, dry rice husk fill)
0.3	2026-09-25	Amish Chadha	Redefine R12 as the cooling equipment kit (ZBX-DDR-001 item 1); R7 on the decided 150 W panel; TRL 3 status from ZBX-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, design point and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

ZeerBox requirements

These requirements are checked by calculation in ZBX-CAL-001 v0.3 at TRL 3, after the design was made constructable (ZBX-DDR-003). Seven are met (three of them by design or logic review only), the kit cost (R12) is USD 1 under its value-engineering target, **R3 is at risk**, none is not met, and R6, R10 and R11 cannot be verified at TRL 3. Targets are not yet validated with users and will be revised after co-design sessions (see ZBX-PRB-001).

The **design point** used throughout is outside air at 35 °C and 30 % RH (wet bulb 21.5 °C), a full store of 480 kg of mixed vegetables, and 100 kg of fresh produce loaded per day at 32 °C.

Table 1. Requirements and TRL 3 status. Values are central estimates from ZBX-CAL-001 v0.3, with the favorable to unfavorable range where it matters.

ID	Requirement	Target	Verification	TRL 3 status (ZBX-CAL-001)
R1	Storage capacity	20 or more standard 20 kg crates (400 kg or more) on shelves; one adult can walk in and reach every crate without lifting any crate above 1.6 m	Layout model; later loading trial	Met: 24 crates, 480 kg; top shelf 1.40 m above the floor
R2	Cooling at the design point	Mean store air temperature 8 K or more below outside air	Psychrometric and heat balance calculation; later logged field data	Met: 10.7 K (9.8 to 11.4 K); store 24.3 °C
R3	Humidity for produce	Store RH 80 % or more while cooling, to limit water loss; covered or lined crates for leafy and water-sensitive produce	Calculation; later logged field data	At risk: 81 % (77 to 85 %)
R4	Behavior in humid weather	Controller detects outside air too humid to cool (wet-bulb depression under 4 K) and switches to night ventilation or hold, with a visible mode indicator	Controller logic review; later bench test with sensors	Met by logic review: 30 °C and 75 % RH gives 3.7 K and switches to hold
R5	Mold and condensation guard	Pad pump stops if store RH stays above 95 % for 60 min; store air changes at least once per hour whenever produce is inside	Controller logic review	Met by logic review: hold mode gives about 8 air changes an hour at about 2 W
R6	Shelf life	Tomato shelf life at least 1.5 times that of ambient storage in the dry season (for example 6 days to 9 days or more)	Literature comparison; later paired storage trial with the partner	Not verifiable at TRL 3; ZECC data (about 1.8 times) come from a cooler, more humid chamber
R7	Solar operation	Fans, pump and controller run from one PV panel with no grid for a full design day: 10 h of cooling plus 2 h of evening ventilation	Energy balance	Met: 372 Wh needed, 525 Wh from the 150 W panel (41 % margin)
R8	Water use	70 L or less of water per design day, including pad and wall cavity; sump holds at least one day	Water balance	Met: about 36 L (no cavity water); 60 L sump for 36 L of pad and bleed water
R9	Low-voltage and safe	All electrical parts at 12 V DC nominal, fused at the battery; fans guarded; door opens from inside without a key	Design review	Met by design review
R10	Local build and repair	Walls, roof, door and shelves built by a local mason and carpenter; every fan, pump, sensor and pad replaceable in 30 min or less with generic parts	Design review; parts availability survey with the partner	Not verifiable at TRL 3
R11	Durability	Structure 10 years or more; pad 3 years or more with cleaning; fans and pump 3 years or more in dust	Material review; supplier data	Not verifiable at TRL 3; roof footings 1.8 times the uplift at 30 m/s, 500 mm kept as a minimum (ZBX-DDR-002); roof beams and purlins sized (ZBX-CAL-001 v0.3)
R12	Affordable			

ID	Requirement	Target	Verification	TRL 3 status (ZBX-CAL-001)
		Cooling equipment kit (pad, sump and pump, fans, controller, solar panel, power box, wiring and panel frames: BOM items 6 to 11, 14 and 18) priced against the USD 300 value-engineering target (<code>project.yaml</code> budget). The structure (items 1 to 5, 12, 15, 16 and 17) is built from local materials and costed separately	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Value-engineering target: USD 300 for the kit. Estimated cost of the constructable design: USD 299 (USD 1 under the target); structure USD 574, reported separately

R12 was redefined on 2026-09-25 by Amish's decision to go with the TRL 2 recommendation (ZBX-DDR-001 item 1). Its earlier target, the complete store for \$300, was not met at about \$575. Since 2026-10-01 the USD 300 figure is a hypothetical value-engineering target, not a limit (Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"), so R12 is reported as over or under the target.

R3 was relaxed from 85 % to 80 % on 2026-09-25 by Amish's decision to go with the TRL 3 recommendation (ZBX-DDR-002, item 11), together with a 150 mm pad and produce-level measures (covered or lined crates). At 85 % R3 would still not be met: the store averages 81 % and 85 % needs a pad about 185 mm deep or more.

■ Assumptions

- Direct evaporative pad effectiveness of 75 % for a 100 mm cellulose pad at 0.6 m/s face velocity, scaled for velocity and depth in ZBX-CAL-001; about 88 % for the decided 150 mm pad at 0.52 m/s. Manufacturer data for deeper pads show 90 % or more ([CELdek specification](#)); 70 to 80 % is assumed for 100 mm.
- The fans are taken as a straight-line curve from 400 m³/h free air to a 50 to 70 Pa shut-off; pad pressure drop is 15 Pa at 1.0 m/s. Both need supplier data.
- The heat load at the design point is 442 W (376 to 558 W), from the heat balance in ZBX-CAL-001 v0.2.
- Dry rice husk in the wall cavity conducts about 0.06 W/(m·K) and is kept dry and pest-free (assumption).
- Dry-season solar resource of about 5 peak sun hours, with 30 % combined losses for heat, dust, PWM charging and wiring.
- Tomato shelf life gains are taken from zero energy cool chamber trials, which run cooler and more humid than a forced-air store; R6 may be optimistic.
- Recommended produce conditions follow the UC Davis Postharvest Center (tomato 12.5 to 15 °C for mature green, 90 to 95 % RH). ZeerBox cannot reach these temperatures in the design case; R2 is set on temperature drop, not absolute temperature.

■ Requirements not met or at risk at TRL 3

- **R3 (humidity), at risk:** the 150 mm pad supplies air at 87 % RH, and the store averages 81 % in the central case but 77 % in the unfavorable case, against 80 %. The air loses about 4 to 5 percentage points of RH per kelvin as it warms across the store. Covered or lined crates hold humidity near the produce; field data will show whether they are enough.
- No requirement is not met. R2 moved from at risk to met (10.7 K, 9.8 K unfavorable) with the 150 mm pad and dry rice husk fill (ZBX-DDR-002, items 11 and 12).